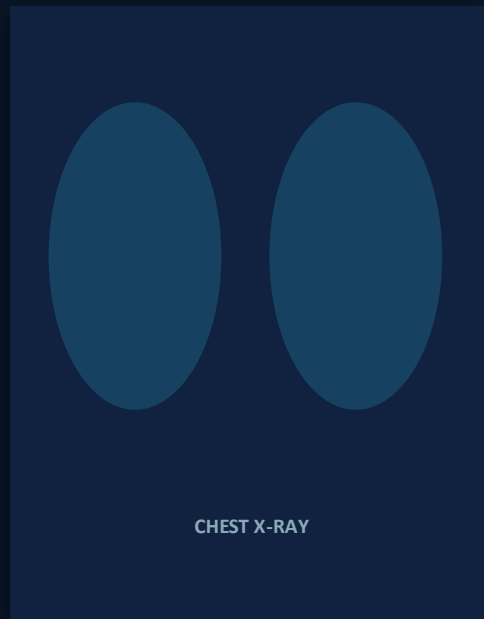


Pneumonia Detection

via Chest X-Ray Classification

Zeinab Drameh · drameh@usc.edu · Spring 2026

CNN · SVM · Random Forest



Problem Statement

Can machine learning reliably distinguish pneumonia from healthy lungs in chest X-rays — and can it support clinicians in making faster, more consistent diagnoses?

Why This Matters

~2.5M deaths/year

Pneumonia causes ~2.5 million deaths annually worldwide — early diagnosis is critical

Radiologist shortage

Many regions lack access to trained radiologists; AI can bridge the gap

Inconsistency risk

Manual reading is time-consuming and subject to inter-reader variability

Speed is life-saving

Faster preliminary screening means faster treatment and better outcomes

The Decision We're Supporting

Who Is This For?

Radiologists

Primary reviewers who interpret X-rays and issue diagnostic reports

Pulmonologists

Specialists who manage respiratory disease and treatment planning

Emergency Physicians

Front-line doctors who need rapid triage of critically ill patients

Primary Care Physicians

General practitioners screening patients before specialist referral

The Core Decision



Flag for Follow-up

Model flags as PNEUMONIA → radiologist confirms and prescribes treatment



Clear Patient

Model outputs NORMAL → physician can deprioritize or conduct further tests



Second Opinion

Low-confidence cases flagged for immediate specialist review

 *Model is a decision-support tool — not a replacement for clinical judgment*

Dataset

~5,800

Total Images

5,216

Training Set

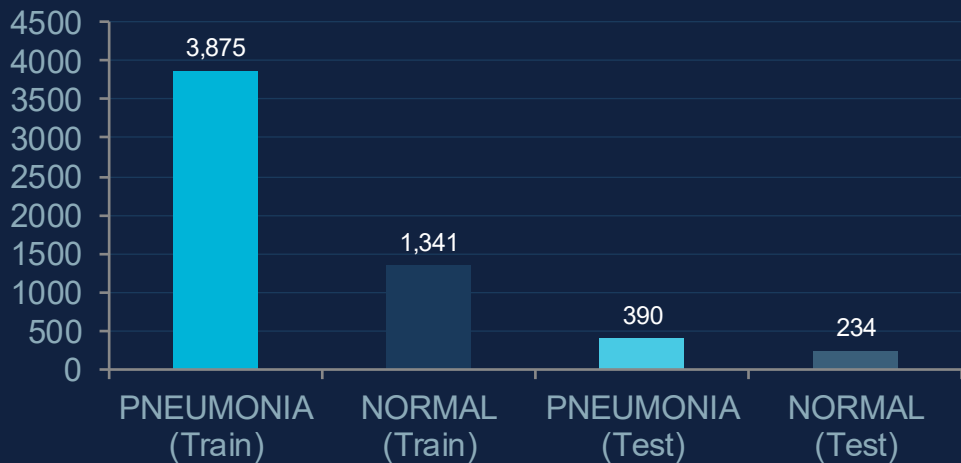
624

Test Set

16

Validation Set

Class Distribution



Key Observations

Class Imbalance

Training data is ~3:1 pneumonia-to-normal. This biases all models toward predicting PNEUMONIA

Tiny Validation Set

Only 16 validation images — unreliable for tuning. Cross-validation was used instead

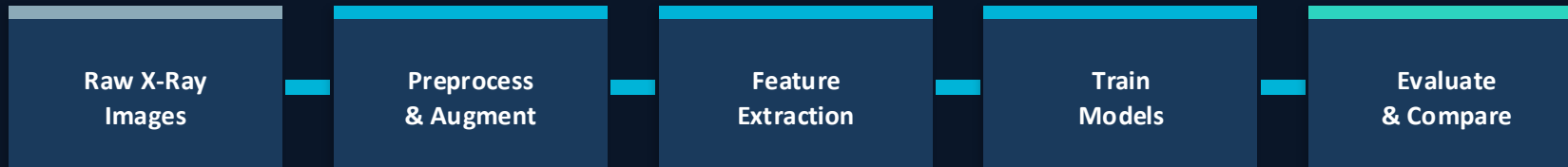
Image Format

Grayscale JPEG X-rays; no tabular features. All signal is in pixel intensities and spatial structure

No Missing Data

No rows removed. Image paths and labels were complete

Methodology Overview



Three Models Compared

CNN	SVM	RF
Convolutional Neural Network	Support Vector Machine	Random Forest
4 conv layers (32→256 filters) 3×3 kernels + max-pooling - Global average pooling - Adam optimizer, BCELoss 20–30 epochs, LR tuned from 0.001→0.01 - Augmentation: crop, flip, resize - Input: 2,000 images (64×64)	4 variants: Linear, Poly+Linear, Poly Kernel, RBF - Images flattened to 1,024-dim vectors (32×32) - StandardScaler normalization - Best: RBF kernel, $\gamma=0.001$, $C=10 \rightarrow 79\%$ - AUC: 0.90 - Subset: 1,000 train / 500 test - No spatial structure preserved	- Ensemble of decision trees 10-fold cross-validation - RandomizedSearchCV tuning - Best params: max_depth=None, min_samples_split=5 - CV accuracy: 93.1% ± 3.4% - AUC: ~0.98 - Subset: 1,000 train / 500 test

Data Preprocessing & Augmentation

CNN Pipeline

1 Random Crop
Introduces spatial variation — forces model to learn partial views

2 Random H-Flip
Doubles effective data with mirrored X-rays

3 Random Resize
Teaches scale invariance

4 Resize to 64×64
Standardizes all inputs for the network

5 Grayscale + ToTensor
Converts to single-channel [0,1] float tensor

6 Normalize [0,1]
Applied uniformly; brightness dropped (altered opacity signal)

SVM & Random Forest Pipeline

1 Grayscale Conversion
Strip color info — X-rays are inherently grayscale

2 Resize to 32×32
Manageable feature vector size (1,024 pixels)

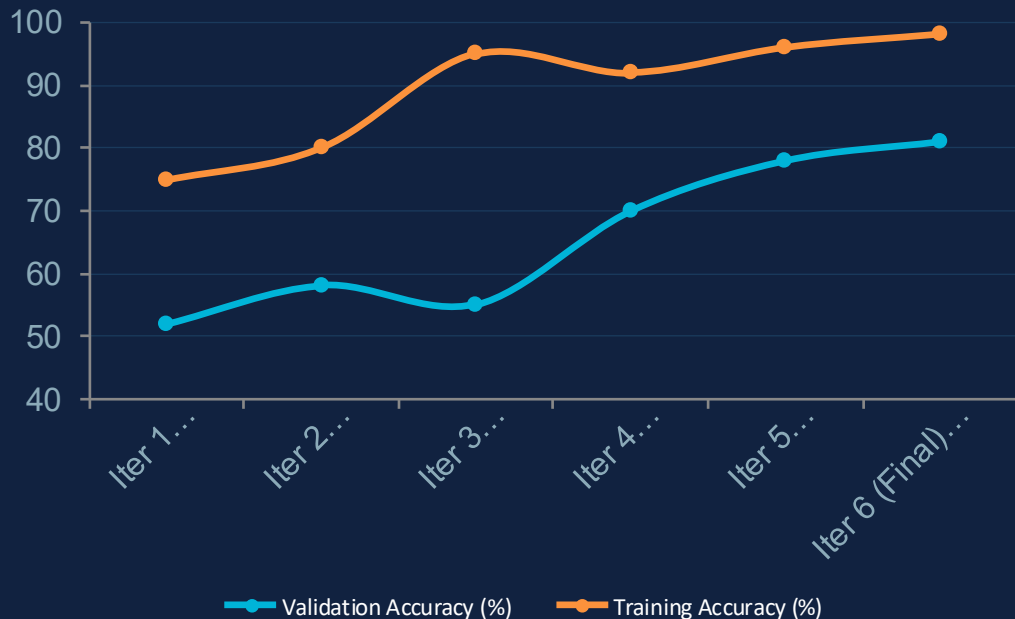
3 Flatten to 1D Vector
Convert 2D pixel grid → single array for sklearn models

4 StandardScaler
Zero mean, unit variance — critical for SVM margin calculation

5 No augmentation
Flipped/cropped images not meaningful for non-spatial models

CNN Hyperparameter Tuning

CNN Training vs Validation Accuracy by Iteration



Key Tuning Decisions

✗ Removed Brightness

Brightness augmentation distorts the opaque areas that distinguish pneumonia — accuracy dropped to 52%

↑ Raised Learning Rate

Increasing LR from 0.001 → 0.01 was the single biggest gain, pushing val accuracy from 58% → 78%

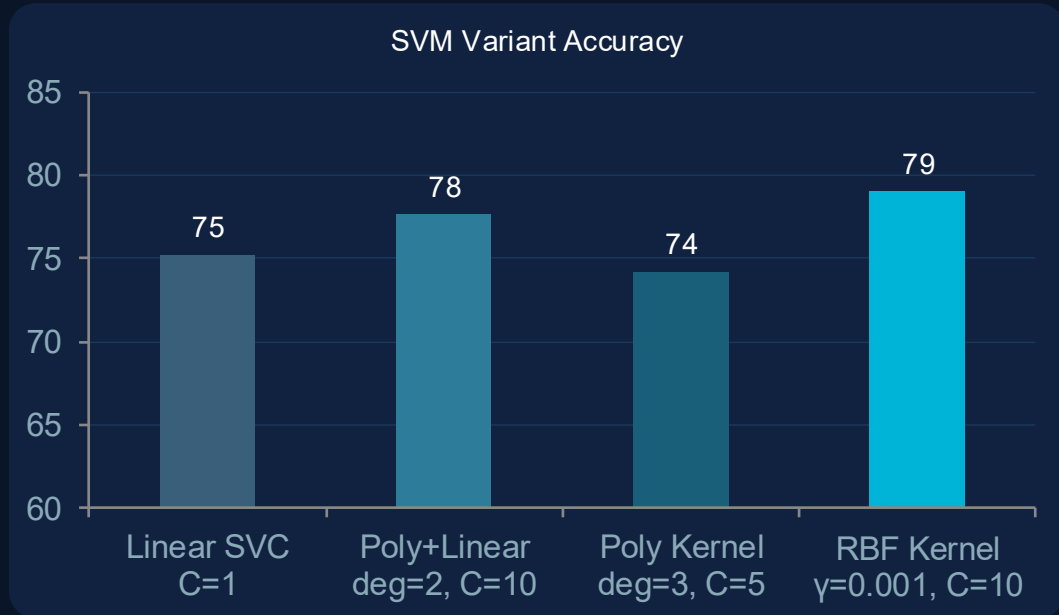
✗ Removed Dropout

With only 2,000 training images, dropout reduced useful signal. Removing it improved training coverage

✂ Cut at Epoch 27

Validation accuracy peaked ~epoch 27 and began to degrade. Early stopping applied to final model

SVM Hyperparameter Tuning



AUC (Best RBF Model): 0.90

Why RBF Wins

Linear SVC (75.2%)

A linear boundary in 1,024-D pixel space captures some signal, but can't model the non-linear patterns in X-rays.

Poly+Linear (77.6%)

Explicit degree-2 features add non-linear combinations. Small but real gain over baseline.

Poly Kernel (74.2%)

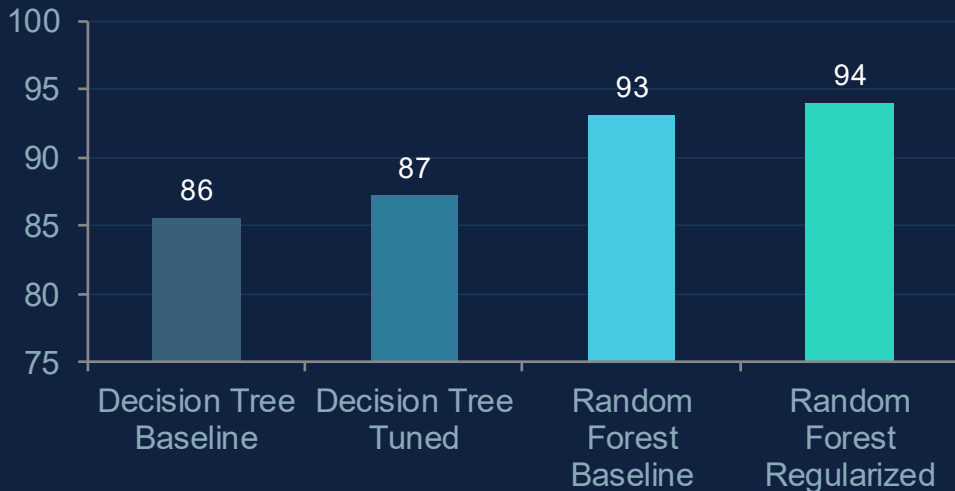
Higher degree (3) without explicit expansion overfit the training data — lower than simpler poly model.

RBF Best (79.0%)

Gaussian kernel measures similarity between examples across the whole feature space. Gamma tuned way down (0.1 $\rightarrow$ 0.001) to prevent memorization.

Random Forest Results

10-Fold Cross-Validation Accuracy



AUC ~0.98

Random Forest ROC

What the Results Show

Why RF beats DT

Ensemble of trees trained on different data subsets reduces variance. Single trees are sensitive to exact training examples.

Best Hyperparams

`max_depth=None`, `min_samples_split=5`, `min_samples_leaf=2`, `criterion=gini`. Controlling leaf size prevents overly specific splits.

Overfitting remains

Training accuracy stays near 100%; validation ~94%. The gap closed after regularization but didn't disappear.

Clinical tradeoff

Confusion matrix: 306/317 pneumonia cases caught (high recall), but 77/183 normals misclassified. In medicine, false negatives are more costly.

Final Test Results

82.4%

Random Forest Accuracy

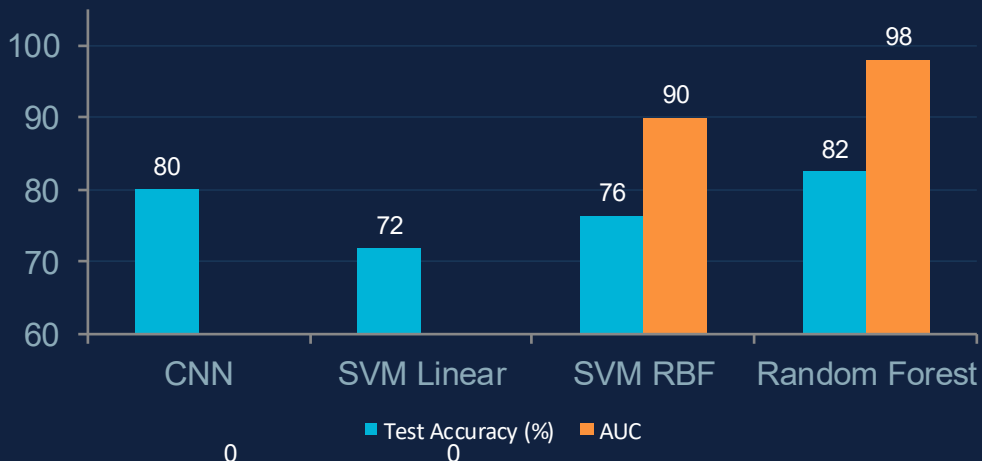
80.0%

CNN Accuracy

76.4%

SVM (RBF) Accuracy

Test Accuracy & AUC (AUC ×100)



Why RF Outperformed CNN

- CNN trained on only 2,000 images (vs. 5,216 available) — underfit to its potential
- RF trained on a representative 1,000-image subset with a solid signal in pixel values
- CNNs need large data to outperform classical methods; spatial advantage didn't materialize at this scale
- RF confusion matrix: 306/317 pneumonia recall — clinically preferable to miss fewer cases
- SVM AUC 0.90 shows well-calibrated probabilities despite lower hard-threshold accuracy

Analysis & Limitations

Class Imbalance

3:1 pneumonia-to-normal ratio pulled all models toward predicting PNEUMONIA, inflating accuracy while masking poor normal-class precision.

Limited Training Data

CNN used only 2,000 of 5,216 available images to keep training time manageable. This directly limited its ability to generalize.

Tiny Validation Set

The 16-image validation set provided by Kaggle was too small for reliable tuning. Cross-validation on training data was used instead.

Spatial Information Loss

SVM and RF flatten images, discarding all spatial structure. They perform surprisingly well but will always have a theoretical ceiling.

Not Clinically Validated

These models were not evaluated against radiologist ground truth or on diverse patient populations. Deployment requires extensive clinical validation.

Brightness Augmentation

Brightness changes distort the opacification signal that distinguishes pneumonia from normal lung tissue. Removed after initial testing.

Note: High overall accuracy does not guarantee clinical safety — precision/recall by class matters more than aggregate accuracy.

Future Recommendations

01 Gather More Data

The model is not ready for deployment. More X-ray images are needed, with a better balance between pneumonia and non-pneumonia cases (closer to 1:1 ratio).

02 Augment Non-Pneumonia 2×

To address the class imbalance, apply augmentation twice as heavily to NORMAL images than to PNEUMONIA images. This creates a more balanced effective training set without requiring new data collection.

03 Transfer Learning (ResNet-18)

Fine-tune a pretrained ResNet-18 on this dataset. Pretrained features from ImageNet generalize well to medical imaging and require far less data than training from scratch.

04 Train CNN on Full Dataset

Use all 5,216 training images rather than a 2,000-image subset. This alone would likely close the gap between CNN and Random Forest performance.

05 Address Class Imbalance Explicitly

Apply SMOTE oversampling or class-weighted loss for CNN; use `class_weight='balanced'` for SVM and RF. Evaluate per-class F1 scores, not just aggregate accuracy.

06 Larger SVM/RF Input

Resize images to 64×64 (4,096 features) instead of 32×32 (1,024) for SVM and RF. More spatial detail should improve their decision boundaries.

Conclusions

Random Forest performed best

93.1% CV accuracy, 82.4% test accuracy, AUC ~0.98 — strongest on this data scale

CNN has higher ceiling, lower floor

Theoretically superior for images; underperformed here due to limited training data and tuning constraints

SVM is a solid baseline

79% accuracy, 0.90 AUC despite using only flattened pixels — interpretable and fast to train

More data is the #1 priority

All models showed overfitting. A larger, balanced dataset is the most impactful next step before any clinical consideration